Inter IIT Tech Meet 15.0 — Sarvam Mini PS

Hindi → English real-time speech-to-speech | Dev-track prototype and data plan

Scope. This report documents protocol tests, a research-led dataset plan and six short recordings. The current server echoes Hindi PCM unchanged; it is not a translator. Transcript drafts and timing estimates are not yet acoustically verified.

1. Problem and evaluation

The task is live translation of naturally code-mixed Hindi to English speech. Hindi SOV order delays the decisive verb; irreversible audio makes premature guesses costly. Names, dates, phone/account digits, amounts in lakh/crore, negation, conditions and self-corrections need explicit protection. KPI weights: translation quality 12, intent accuracy 18, entity accuracy 18, latency 16, audio quality 6. Final gates cover interface/container, fine-tuning evidence and a valid frozen dataset; any gate failure zeros final KPI points.

2. Streaming protocol

Accept WebSocket at **:8000/ws**; **GET /health** must return HTTP 200 on port 8000. Receive session_start JSON, then mono PCM16 24 kHz input in 80 ms/3,840-byte frames, then input_end with valid_samples. Keep reading input while output is paced; the harness never waits. Finish within 30 seconds of input_end, emit output frames ≤1 second, send exactly one session_complete and no later messages. Accept a connection within 10 seconds.

3. Correction policy

For “■■■■ ■■■… ■■■■ ■■■■, ■■■■ ■■■■■”, transcript the whole utterance but translate the final corrected intention only. A risk-aware policy could delay committing output around amounts, names, negation and correction cues, while streaming lower-risk clauses. This is a research hypothesis, not a measured gain.

4. Streaming research

STACL (Ma et al., 2019) provides a reproducible wait-k prefix-to-prefix baseline. Information-Transport-based Policy (Zhang & Feng, 2022) motivates adaptive read/write decisions. SimulEval (Ma et al., 2020) provides a simultaneous evaluation framework. AL can be biased by output-length mismatch; LAAL (Papi et al., 2022) should be reported alongside AL. Test k=2/3/4 against correction-aware buffering using the same references and report quality, AL/LAAL and entity/intent preservation jointly.

5. Dataset design

Proposed 200-hour ceiling (not claimed collected): 100 h licensed real speech only after redistribution rights are confirmed; 50 h consented volunteer speech; 50 h synthetic Hindi with model/voice/version logged. Start with a 6–10 h pilot. Split 80/10/10 by speaker/source to prevent leakage. Keep clips ≤60 s and de-duplicate overlapping real source intervals.

Reproducible pipeline

Licence/consent + checksums → mono PCM16 24 kHz conversion → verbatim transcript → human correction → word-level alignment → literal/fluent English translation → entity/intent checks → schema validation → immutable JSONL manifest. Validate ordered non-negative timestamps within clip duration, complete source metadata and stable IDs. For post-midterm English supervised targets, use Bulbul v3, voice shubh, en-IN, 24 kHz; archive settings and align English words separately. No external ASR/MT/LLM/TTS cascade may steer final inference.

Quality risks

Code-switch normalization, number/entity corruption, negation loss, correction reversal, alignment drift, speaker leakage, synthetic-speaker bias and licence uncertainty. IndicVoices demonstrates breadth across Indian speech, but each subset licence and domain must still be checked. MuST-C demonstrates aligned speech/transcript/translation data; FLEURS offers multilingual evaluation but does not replace spontaneous code-mixed testing.

Four cheap experiments

1) Wait-k sweep k=2/3/4 with AL+LAAL. 2) 100-item entity stress set for names, amounts, dates, phones, negation and corrections. 3) Clean/noisy matched evaluation with noise augmentation applied during training only. 4) Human boundary audit of 30+ transitions per clip; report median and 95th percentile absolute timing error.

6. Six clips and annotation status

Six user-provided WAVs are mono, 24 kHz and 23.04–30.87 seconds each. The repository contains transcript/translation drafts derived from the intended scripts and a word-timing initialization based on token duration weights. It is **not** forced alignment and is **not** hand-checked. The boundary audit sheet contains sampled transitions marked pending. Entity references are listed, but model entity accuracy is null because no translated output exists.

7. Dev-track protocol results

Test	Frames in/out	First output	Wall time	Result
Paced client	382 / 382	0.232 s	35.686 s	1 completion; no later messages
Fast client	382 / 382	0.230 s	30.720 s	1 completion; no later messages
Fast + explicit URI	382 / 382	0.235 s	30.728 s	1 completion; no later messages
Fast + explicit URI	382 / 382	0.233 s	30.719 s	1 completion; no later messages

Health endpoint

User-reported `curl.exe -i http://localhost:8000/health` returned HTTP/1.1 200 OK with body `{"status":"ok"}` and server `Python/3.11 websockets/12.0`.

Interpretation

Basic Docker health and echo protocol tests passed in the user-reported environment. Equal frame counts are expected for echo. First-output timing around 0.23 s is not translation latency; no translation, intent, entity or audio quality score was measured. Disconnect, malformed-frame and backpressure stress tests remain outstanding.

8. Limitations and honest status

The package does not include a translated checkpoint, fine-tuned weights, training logs, Bulbul target pairs or before/after model scores. Transcripts, translations and timestamps must be checked against the actual audio before being called final. At least 30 boundaries must genuinely be listened to and corrected; the audit sheet is pending, not complete. Public data must not be redistributed until licence terms are verified.

Recommendation

Present this as a protocol/data-pipeline prototype with transparent limitations. Do not claim it meets the full translation task or all final gates.

References

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